Algorithm

Algorithm 1: Add Student

1. Start.
2. Input roll number.
3. Check whether the roll number already exists.
4. If it exists, display a duplicate-record message and stop this operation.
5. Otherwise input student details, subjects, marks, attendance and contact information.
6. Store registration number and date of birth as a tuple.
7. Store the complete student profile in a dictionary using the roll number.
8. Display successful message.
9. Stop.

Algorithm 2: Search Student

1. Start.
2. Input roll number.
3. Search the main dictionary.
4. If found, display the complete record; otherwise display Student not found.
5. Stop.

Algorithm 3: Update Student

1. Start.
2. Input roll number.
3. Check whether the student exists.
4. Display update options.
5. Update the selected field.
6. Display success message.
7. Stop.

Algorithm 4: Delete Student

1. Start.
2. Input roll number.
3. Check whether the student exists.
4. Delete the record if found.

5. Display result.

6. Stop.

Algorithm 5: Calculate Average Marks

1. Start.

2. Input roll number.

3. Retrieve the marks list.

4. Calculate sum of marks divided by number of marks.

5. Display average.

6. Stop.

Algorithm 6: Find Highest Scorer

1. Start.

2. Check that records exist.

3. Calculate each student's average.

4. Compare averages and retain the highest.

5. Display the highest scorer.

6. Stop.

Algorithm 7: List Students by Department

1. Start.

2. Input department.

3. Check every student record.

4. Display students whose department matches.

5. If none match, display an appropriate message.

6. Stop.

Algorithm 8: Count Students

1. Start.

2. Find the length of the students dictionary.

3. Display the total.

4. Stop.